

Report
Of
AI-ML Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student:2410031410

Name of Student: NAINA

Batch: 2024-28

2026-27

CANDIDATE'S DECLARATION

I, Naina do hereby declare that the internship report titled “**AI-ML Virtual Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Naina

2410031410

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Naina

2410031410

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



एन सी ई आर
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Naina

IILM University, Greater Noida

has successfully completed the 8-weeks

AI-ML Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : India Edu Program

Google for Developers



Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4c268b22cc7452536909
Student ID: 218161443126



GRADE: O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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4. PROJECT DESCRIPTION

4.1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally. Early risk stratification can significantly improve clinical outcomes. This project develops an end-to-end Machine Learning Web Application that assesses a patient's risk of heart failure based on 11 clinical attributes.

4.2 Organization Profile

The internship was supported by AICTE, EduSkills, and Google for Developers under the AI-ML Virtual Internship program (April - June 2026).

4.3 Problem Statement

Traditional diagnostic workflows often rely on static clinical guidelines that can obscure complex non-linear feature interactions across multiple cardiac indicators.

4.4 Project Objectives

The primary objectives are to train a supervised Machine Learning classification model to predict heart failure probabilities, implement data preprocessing (One-Hot Encoding and Standard Scaling), and build a Flask application to serve model predictions via a responsive user interface.

4.5 Scope of the Project

The project scope is limited to utilizing the Kaggle Heart Failure Prediction Dataset with 918 observations across 11 diagnostic variables. The application categorizes patient risk into three distinct tiers: Safe, Serious, and Critical.

4.6 Technologies and Tools Used

The system is developed using Python, scikit-learn, Pandas, NumPy, and Joblib for machine learning. The backend is powered by Flask, while the frontend interface utilizes HTML5, Tailwind CSS, JavaScript, and an HTML Canvas Gauge.

4.7 System Architecture

The system architecture consists of a Flask Web Server that handles form input parsing, applies feature reindexing, standardizes features using a loaded scaler, and outputs prediction probabilities to a Tailwind CSS frontend with a custom SVG gauge meter.

4.8 Methodology

Categorical features were dummy-encoded via One-Hot Encoding, while numeric features were scaled using StandardScaler inside a ColumnTransformer. A Logistic Regression classifier was fitted, and the resulting pipeline was serialized into .joblib files for real-time inference.

4.9 Expected Outcomes

An interactive web interface that provides instant visual feedback categorizing risk into Safe ($<30\%$), Serious ($30-70\%$), or Critical ($\geq 70\%$) alongside automated medical advice generation.

4.10 Certificates of completion and other communication mail proof must be attached

The official Internship Completion Certificate (ID: 4c268b22cc7452536909) demonstrating Grade-O (Outstanding) performance is formally attached to this document as per the submission requirements.

5.BIBLIOGRAPHY/REFERENCES

1. Kaggle Dataset: Heart Failure Prediction Dataset (heart.csv).
2. Scikit-learn Documentation for Logistic Regression and Preprocessing pipelines.
3. Flask Framework Official Documentation for Python Web Servers.